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RESEARCH ARTICLE

A Transparent Search-Based Framework for Skin and Breast Cancer Diagnosis Using Siamese Networks

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ABSTRACT Computer aid diagnosis has developed digital pathology with Deep Learning (DL)-based tools to assist pathologists in decision-making. Content-Based Histopathological Image Retrieval (CBHIR) is a novel tool to seek highly correlated patches in terms of similarity in histopathological features. In this work, we proposed two CBHIR approaches on breast (Breast-twins) and skin cancer (Skin-twins) datasets for robust and accurate patch-level retrieval, integrating a custom-built Siamese network as a feature extractor. The proposed Siamese network generalizes for unseen images by focusing on the similar histopathological features of the input pairs. The proposed CBHIR approaches are evaluated on the Breast (public) and Skin (private) datasets with top K accuracy. Finding the optimum amount of K is challenging; also, as much as K increases, the dissimilarity between the query and the returned images increases, which might mislead the pathologists. This paper is tackling this issue for the first time on histopathological images by evaluating the top first retrieved images, while such approaches are underexplored in existing literature. The Breast-twins model achieves 70% of the F1-score at the top first, which exceeds the other recent methods at a higher amount of K such as 5 and 400. Skin-twins overpasses the recently proposed Convolutional Auto Encoder by 67%, increasing the precision. Besides, the Skin-twins model tackles the challenges of Spitzoid Tumors of Uncertain Malignant Potential (STUMP) to assist pathologists with retrieving top K images and their corresponding labels. So, this approach can offer a more explainable tool to pathologists in terms of transparency, trustworthiness, or reliability.

INDEX TERMS Siamese network, computer-aided diagnosis (CAD), digital pathology, histopathological images, statistical learning, content-based histopathological image retrieval (CBHIR).

I. INTRODUCTION

Skin cancer is one out of three diagnosed cancers worldwide, according to the World Health Organization (WHO) [1]. Basal and squamous cell carcinoma are the two most frequently occurring types of skin cancer, while the most perilous type is malignant [2]. Classifying and distinguishing between benign (melanocytic nevus) and malignant (melanoma) can be reliably feasible in common melanocytic tumors. One of the diagnostic challenges for pathologists in spitzoid tumors is Spitzoid Tumors of Uncertain Malignant Potential (STUMP) because their prognostic implications are unknown [3]. In this case, in order to make an accurate cancer diagnosis, pathologists often need to transfer the glass of

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biopsy to other centers to consult with their peers about the grade of the tissue. This workflow is time-consuming and expensive, and some accidents might occur with the glass of the biopsy [4].

Another world's most prevalent cancer type according to WHO is breast cancer with 685, 000 deaths globally in 2020 [5]. Breast cancer is caused by abnormal breast cells that grow out of control and form tumors. Cancer cells can spread to nearby lymph nodes or other organs. Treatment for breast cancer depends on the type and sub-type of cancer and how it has spread outside of the breast [1]. There are some treatments for breast cancer, but overdiagnosis or overtreatment and variability in interpretation are some key challenges in breast cancer diagnosis [6].

Computer Aid Diagnosis (CAD) offers some efficient Deep Learning (DL) tools that can assist pathologists and address human errors in cancer diagnosis and treatment. The advance of DL methods opened the door to reducing the workload for pathologists and enhanced patient care [7]. Digitization of the tissue slides as high-resolution Whole Slide Images (WSIs), emerges computer vision tools as a support for pathologists in their daily tasks. Some of these tools, such as classification [8], segmentation [9], content-based Histopathological Image Retrieval (CBHIR) [10], etc., provide a second opinion for pathologists to have a more accurate cancer diagnosis [11]. Image-based tools of CAD are classified into medical image classification and Content-Based Medical Image Retrieval (CBMIR) [12]. The main objective of the classifiers is to categorize the images, while CBMIR aims to rank and show similar images in addition to their labels [13]. CBMIR, particularly CBHIR tools, demand representative features to retrieve similar histological patches with the same histopathological patterns effectively [14]. To reach the most meaningful features of the patches, a proper feature extractor is needed. Finding an optimum feature extractor for the goal under study is challenging. There are plenty of FE, including SIFT [15], and Gabor features [6], as well as some handcraft features. In the other works [16], DL-based FEs, including CNN, pre-trained models, Auto Encoder (AE), etc., were applied to extract the deep features.

As a means to tackle this lack of enough annotated images, Zahra et al. proposed an unsupervised ResCAE to perform a search engine in prostate cancer. This paper reported 85%, 78% of accuracy at top 7 and 5, respectively [17]. As another unsupervised technique, in [6], the Gabor feature was used to characterize the texture information, and a local statistical feature of nuclei is presented for nuclei's morphology and distribution. The framework in [18] is fully based on kernel techniques, which can handle a variety of visual descriptors and create expressive feature spaces. Using the query-by-example paradigm, the created annotations were utilized to search for images in an image retrieval system that have similar annotations.

The authors of [19] developed a scalable framework for picture retrieval based on the supervised hashing technique, and they verified its efficacy using thousands of histopathology images taken from microscopic breast tissues. The proposed method in [19], achieved about 88.1% classification accuracy. Histopathology Siamese Deep Hashing (HSDH) is proposed in [20] for histopathology retrieval and achieved 97%, 98%, and 99% Mean Average Precision (MAP) for 32, 64, and 128 bits. The authors reported the results at the top 100, 150,..., and 400 for BreaKHis as a binary data set. RetCCL in [21] is a clustering-guided contrastive learning approach for WSI retrieval. RetCCL has improved 24% in patch-level retrieval on the TissueNet data set in terms of $mMV@5$ compared to ImageNet pre-trained features. The authors in [22] proposed a CBHIR framework for a data set that contains WSIs and scalable query regions of interest (ROI) of size. The reported retrieval results at the top 20 returned images reached 96% of precision on the Motic database. SMILY [23] was proposed by Google AI Healthcare, and it used an automatic high-level feature extraction to provide the feature vectors for the search engine. SMILY retrieved images with the top 5 similar patches of prostate cancer with 73% accuracy.

In the recent papers on CBHIR [4], [17], [20], [22], [23], [24], [25], the performance of the model was reported by top K accuracy. This evaluation technique assumes that if just one of the retrieved images out of K is correct, the model performed correctly [21]. Choosing the correct amount of K in this evaluation technique is challenging. In some papers, such as [20], the amount of K was chosen in $[100 - 400]$, which is a high amount, especially for a binary data set. In [22], the reported accuracy is at the top 20 images. In other papers [26], [27], mostly the amount of K was chosen as $5, 7, 10$. Retrieving top K images brings some benefits for pathologists; for instance, they can analyze more cases to see similar histopathological features. The issue with top K accuracy in evaluating the performance of CBHIR is that, in some cases, the model could retrieve one correct similar patch out of 5, 7, or even 400 retrieved images. Although this can ensure pathologists that at least one image out of K is correct, it cannot provide a highly accurate second opinion for them. However, in the top K accuracy technique, even if the k_{th} retrieved out of K was the only similar image to the query, the accuracy of the model is high. This issue is highlighted more in the papers with a high amount of K. In contrast, retrieving the most relevant image at top-1 is particularly valuable in clinical practice, where a precise and immediate match can support faster, more confident diagnostic decisions and reduce ambiguity in expert assessment.

Departing from conventional approaches, this paper proposes a CBHIR system that utilizes a custom-built Siamese network as a feature extractor for skin and breast cancer datasets—two of the most prevalent cancer types. Given the significant variability within these datasets, the Siamese network offers a promising solution for analyzing histopathological images. It is specifically adapted to capture the complex patterns in image patches that are essential for accurate grading and diagnosis.

The Siamese network excels in similarity-based tasks, making it particularly well-suited for CBHIR applications. By employing a contrastive loss function, the network emphasizes relationships between data points in the embedding space, enhancing its ability to identify subtle similarities between images while effectively distinguishing dissimilar ones. The features extracted by this network are highly transferable and can be integrated into the CBHIR search engine without requiring extensive retraining—an important advantage when working with histopathological images. Moreover, this approach addresses common challenges such as small inter-class variations and large intra-class differences.

In this work, we introduce two Siamese network-based models tailored specifically for breast and skin datasets, named Breast-twins and Skin-twins. While both models are built upon the Siamese architecture, they differ in their configurations to better accommodate the unique characteristics of each dataset. The following sections provide detailed descriptions of each model's architecture, training process, and hyperparameter tuning. Figure 1 illustrates the proposed CBHIR system in a general overview. In summary, this paper makes the following main contributions:

1) We propose two Siamese networks for breast cancer (public data set) and skin cancer (private data set) to show the generalization of the proposed CBHIR technique. These proposed Siamese networks are robust to imbalanced data sets;
2) To the best of the author's knowledge, no previous studies were conducted based on the Siamese network in a CBHIR model to assist pathologists in making a more accurate diagnosis of spitzoid cancer.
3) The CBHIR results on both data sets were reported at the top first retrieved images to demonstrate the model's efficacy in retrieving relevant patches;
4) The proposed CBHIR model can have high performance with the top first retrieved patches, which is the first paper that could reach this power at the top first, to the author's best belief. The performance of the proposed CBHIR model confirms that the model is generalized well to unseen data;
5) The proposed CBHIR model has been dedicated to grading the STUMP cases in a spitzoid melanocytic lesions database, which remain relatively unexplored in current research;
6) The proposed CBHIR approach provides a second opinion to pathologists to tackle the challenges in grading STUMP by providing deep insights into the complexities. We show the Gradient-weighted Class Activation Mapping (Grad-CAM) figures as explainable Skin-twins to provide interpretability to the uncifrable STUMP cases.

II. METHODOLOGY

First, medical centers have to scan the biopsies with WSI scanners. Second, the Siamese network needs to train to extract the features of the images. Third, the extracted features of the database have to be indexed and saved. Later, a search process needs to find similar patches to the query among the images in the database. Finally, it is time to visualize the top similar patches to pathologists for further analysis. This is how the CBHIR system can make a bridge between the current cancer diagnosis and digital pathology.

In this paper, we proposed two models based on a Siamese network for breast and skin data sets, which are named Breast-twins and Skin-twins. Both Breast-twins and Skin-twins were designed to be lightweight and efficient, using shallow architectures and compact embeddings that enable fast feature extraction and retrieval.

A. SIAMESE NETWORK

The Siamese network is characterized by its twin neural network structure. Siamese networks are designed to compare the similarity between pairs of inputs. By training with image pairings, Siamese networks, concentrate on similarity learning and produce a distance metric as a result of learning a feature space for image similarity. In comparison with Convolutional Neuron Networks (CNNs), Siamese networks are better at precisely locating related images based on feature distances. This network can be effective even with limited images in the train set, and it learns the representative features by focusing on the relative relationships between the pairs of input. Siamese networks bring advantages to the CBHIR approach, such as robustness to class imbalance and learning from semantic similarity.

Figure 2[1] As can be seen, there are two identical networks, which are named sister networks. Each of these sisters contains the same architecture comprising a deep convolutional network, an encoding layer, and a distance metric. The main objective of these sisters is to minimize the contrastive loss in order to minimize the distance between images with the same histological features.

Each sister network takes one instance from the input pair and passes it through the convolutional layers of the proposed network for feature extraction. While training, these sisters shared the weights regarding encoding the input images to ensure that both networks learn to extract similar features of the input image. During training, in order to measure if each pair is similar or dissimilar, the label of the data is needed. Once the Siamese network learns to extract the representative features for pairs of input, it can be used to measure the similarity between pairs by computing the distance between the extracted features generated by the sister network for each patch instance.

B. CONTRASTIVE LOSS

Contrastive loss, compared with the other loss functions, enables the network to converge faster and reduces

[1]The figure is plotted by breast histopathological images. The structure of these convolutional networks is slightly different for skin and breast data sets regarding filter size due to the difference between the size of images in each data set. More details are mentioned in section III-B2.

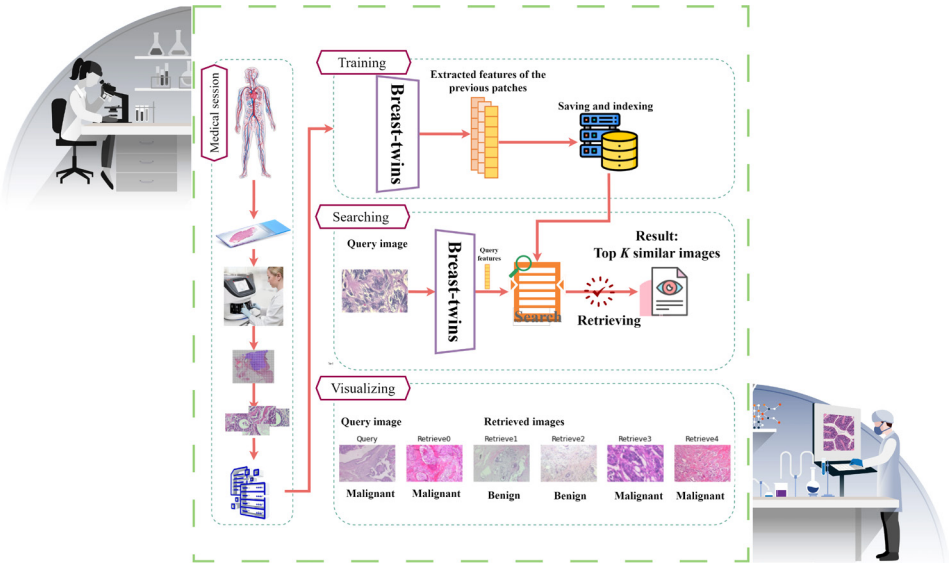


FIGURE 1. A general overview of the proposed CBHIR to show how the proposed CBHIR can bridge traditional cancer diagnosis workflow to digital pathology [4].

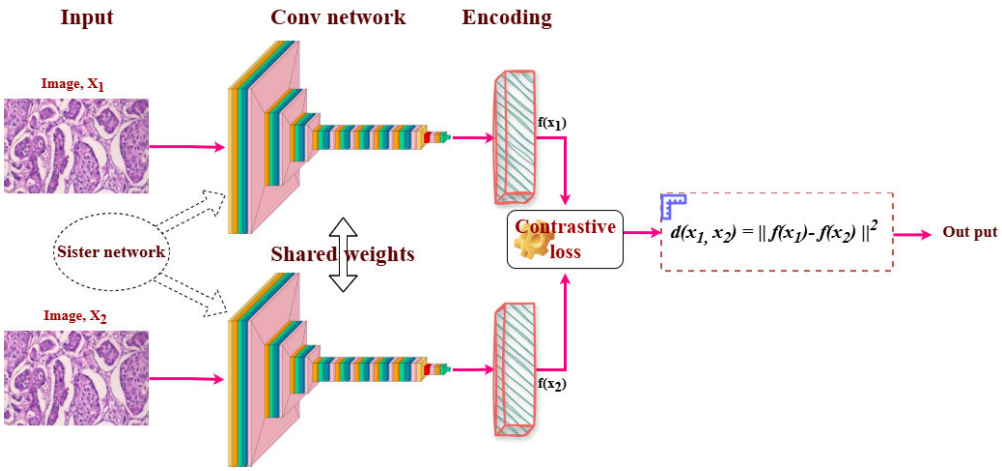


FIGURE 2. An overview of the proposed network. This is plotted with BreaKHis images. Colors of orange, pink, red, green, and teal represent Convolutional layers, Dropout, MaxPooling2D, Dense, and Flatten, respectively.

overfitting. This occurs because the network learns to be more generalized by focusing on the key features and determining their similarity.

The contrastive loss function was employed as it directly optimizes the embedding space for similarity learning, aligning naturally with the goals of CBHIR. Compared to alternatives such as triplet or cross-entropy losses, contrastive loss offers greater simplicity, stability, and proven effectiveness in Siamese-based retrieval frameworks.

The contrastive loss takes pairs of samples as anchor and neighbor or anchor and distant [28]. In the case of anchor and neighbor, they are pulled towards each other. In the other case, the distance between the anchor and the distant needs to be increased. The optimum embedding feature space is extracted while the anchor-distant distances get larger than the anchor-neighbor distances by a margin of m. This can enhance the embedding space and improve the discriminative capacity. The utilized contrastive loss defined for this work is as follows:

$$L_c = \sum_{i=1}^{b} [(1-y)||f(x_1^i) - f(x_2^i)||_2^2 + y[-||f(x_1^i) - f(x_2^i)||_2^2 + m]_+] \quad (1)$$

The contrastive loss (L_c) (1) should be minimized, which helps the network focus on capturing invariant features [28]. y is zero or one when the pair $\{x_1^i, x_2^i\}$ is anchor-neighbor and anchor-distant, respectively. We denote the m as the margin and $f(x)$ as the output of the network (i.e., embedding). Let b be the mini-batch size, and $||.||_2$ the L_2 norm [29]. Figure 3 illustrates the workflow of the contrastive loss by receiving anchor-neighbor and anchor-distant. Three points

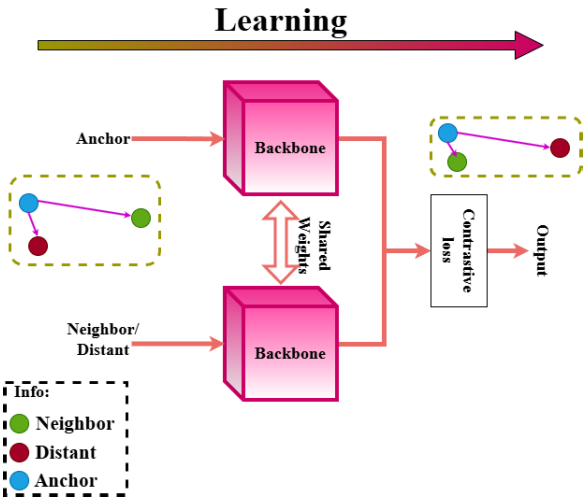


FIGURE 3. An overview of how contrastive loss receives the anchor-distant and the anchor-neighbor. The blue, red, and green points are anchor, distant, and neighbor samples.

in the colors red, green, and blue correspond to distant, neighbor, and anchor samples. Initially, the green point is located far from the anchor sample. As the model undergoes training, it acquires the ability to discriminate between samples. Ultimately, the contrastive loss function propels the red point to a distant location while maintaining the green point close to the anchor sample.

C. SEARCH ENGINE

The trained Siamese network provides a feature extractor that can extract the meaningful features of the data set. Figure 4[2] shows an overview of the proposed CBHIR. Each image i in the data set is saved with a feature vector F_i containing the two most representative features of that. A constructive loss function is then used to compare the encoded features, determining the separation between them. The distance metric, $d(X_1, X_2)$, shows how the encoded features of the two input images differ or are similar. For comparable images, the objective is to minimize this distance, and for different images, to maximize it.

All these feature vectors are indexed and saved in the feature storage. By saving the encoded features and properly indexing them, the system can quickly compare new query images to the stored features, significantly speeding up the search and retrieval process.

When the CBHIR model receives a new query, the feature extractor extracts the most meaningful features of the query (F_Q). As the next step, a distance function is needed to measure the distance of F_Q with all saved F_is in the feature storage (F_D). Among different types of distance functions, Euclidean distance was chosen for this paper as it is one of the most used metrics to calculate the distances between two images in CBHIR. In the searching part of the retrieval

[2]The figure shows the histopathological breast images. The same pipeline was applied to the skin images.

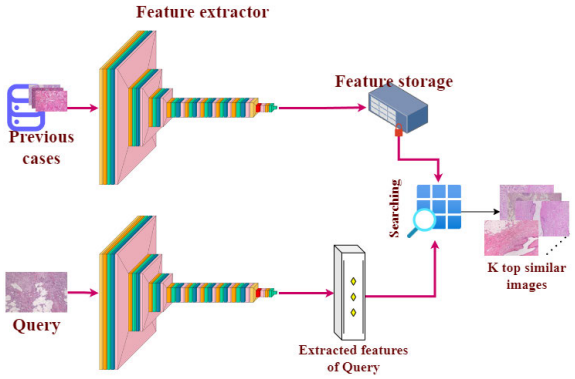


FIGURE 4. An overview of the proposed CBHIR by using the well-trained Siamese network as the FE.

step, the images are ranked with the shortest distance at the top, based on the Euclidean distance. Then, top K retrieved images are returned to pathologists for further investigations. So, it is expected to have the most similar images at the first top retrieved and the least similar image at the K_{th} ($K > 1$) retrieved. As much as the amount of K is less, the number of images that are taken under evaluation is less. So, the model that reaches higher accuracy in a smaller number of retrieved images is a more reliable tool. In this paper, the top 1, 3, and 5 retrieved images on both data sets are shown.

III. MATERIAL AND EXPERIMENTAL SETUP

A. MATERIAL

Two binary data sets of breast cancer and skin cancer were taken under study in this paper. Both data sets were stained with Hematoxylin and Eosin (H & E) [30], [31].

1) BREAST CANCER DATA SET

BreaKHis data set contains 7909 microscopic images of breast tumors at four magnification levels. This paper applied the experiments to images at 400× magnification of this dataset in the size of 224 × 224 × 3 [24]. In grading breast cancer, the number of mitoses is a critical criterion. In order to measure it, 400× magnification was selected, which is the highest level of magnification of BreaKHis. Working at this high magnification provides greater cellular detail and allows more accurate identification of mitotic figures, which is essential for reliable cancer grading. Moreover, the 400× magnification level is the most commonly used setting in prior studies using the BreaKHis dataset, making comparisons with existing literature more fair and consistent. This alignment ensures that experimental results are both robust and comparable to state-of-the-art benchmarks.

This binary data set comprises 1232 malignant and 588 benign at 400× magnification [4]. To ensure a fair and reproducible evaluation, we adopted the same fixed data split and evaluation protocol as prior CBHIR studies on the BreaKHis dataset, allowing consistent comparison with existing methods.

TABLE 1. Description of the Skin data set. This table details the number of whole slide images (WSIs), patients, annotated, and unannotated cases categorized as malignant and benign.

Data type	Malignant	Benign
WSIs	34	50
Patients	30	49
Annotated	16	11
Unannotated	29	41

2) SPITZOID MELANOCYTIC DATA SET

The data set was collected from 79 different patients, containing 84 biopsies. Table 1 describes the spitzoid melanocytic lesions database [32]. The data set is collected and annotated by the pathology laboratory from the University Clinic Hospital of Valencia. The data set contains uncommon skin tumors and 42 STUMP images.

Following [32], the images were segmented from ROIs. The segmented images are non-overlapped, in the size of $512 \times 512 \times 3$. Following [7], in our experiments, in order to prevent data leakage, the data set was later split into 70% training. In this paper, in the training, validation, and testing steps, the STUMP images were excluded. Later, in Section IV-C, they are considered as the second test set, and the trained CBHIR is applied to them to retrieve the most similar patches. In this experiment, we can elucidate if the STUMP query provides more malignant or benign retrievals.

B. EXPERIMENTAL SETTING

The proposed convolutional network in this sister network comprises two stages of convolutional layers. More details about the Siamese structure and the hyperparameters for the breast (Breast-twins) and skin (Skin-twins) data sets are mentioned in the following subsections and Figure 5. The residual blocks shown in Figure 5, highlighted by green dashed boxes, comprise several convolutional layers that contribute to deeper feature extraction. These blocks enhance the model's ability to learn complex representations by adding the input to the output of the convolutional layers, thereby improving gradient flow and stability. The Siamese architecture was deliberately designed as a lightweight, custom-built network that employs optimized residual blocks to achieve a balance between interpretability, computational efficiency, and retrieval performance, while also reducing the risk of overfitting on small-scale medical datasets.

Tuning hyperparameters in Siamese networks is challenging for different purposes, and it relates to the type of the data set. So, it is important to note that Breast-twins and Skin-twins models were trained with a learning rate schedule from Keras with an initial learning rate of 0.01, decay steps of 10000, and decay rate of 0.9. Stochastic Gradient Descent (SGD) optimized the contrastive loss while training. The hyperparameters were tuned empirically within literature-informed ranges, guided by prior studies, and validated based on training stability and retrieval performance. For both datasets, the margin parameter (m) was empirically tuned within a bounded range based on prior studies and validation performance, with m = 0.9 identified as the optimal value that provided the most stable convergence and highest retrieval accuracy.

First, the network is initialized with a contrastive loss and SGD as the optimizer. Then, the first image of the image pairs is processed through the network. Subsequently, the second image of the image pair is fed into the network. Then, the loss is determined by the outputs from the first and the second images.

The experiments in this paper were run on GPU with the *NVIDIA GeForce RTX 3090*.

1) BREAST-TWINS IN DETAILS

Breast-twins model start with four convolutional layers [32, 64, 128, 256] with the filter size of 3 by 3 while stride is [1, 2, 2, 2]. In order to extract deeper features, the network goes deeper with more convolutional layers with the stride of 1 and a kernel size of 3 by 3. The number of convolutional filters in each layer of the residual block is [64, 32, 1, 256] to reduce the spatial dimensions while increasing the number of filters and compressing the information. In this case, decreasing the filter size helped control the number of parameters while still capturing relevant features. Ultimately, a convolutional layer and a Global Max Pooling2D (GMP) are applied to reduce the spatial dimensions and retain important features.

In this experiment, the Breast-twins network was trained over 300 epochs with a batch size of 16. In order to tune and validate the Breast-twins better, we divided the data set into 3 parts including 70% training (4704 patches), 10% testing (655 patches), and 20% validation (1177 patches).

2) SKIN-TWINS IN DETAILS

reflect benign

The Skin-twins network was trained over 300 epochs with a batch size of 16. We divided the data set into three sections for training (70%), validation (20%), and testing (10%) in order to train Skin-twins. Figure 5 provides an overview of the proposed structure from the convolutional point of view.

In the training part of the Skin-twins, the skin data set is considered a binary data set and excludes the STUMP class of the data set to use as the test set for the experiments in section IV-C.

IV. RESULTS AND DISCUSSION

This section reports the results of Breast-twins and Skin-twins with tables of the results and visual evaluations to provide a comprehensive comparison. In this study, the evaluation was conducted on held-out portions of the same datasets to maintain consistency with previous CBHIR studies and ensure fair comparison with existing methods. The train/test set were completely independent. We acknowledge that testing on an independent external dataset would provide a stronger validation of generalization, and we have highlighted

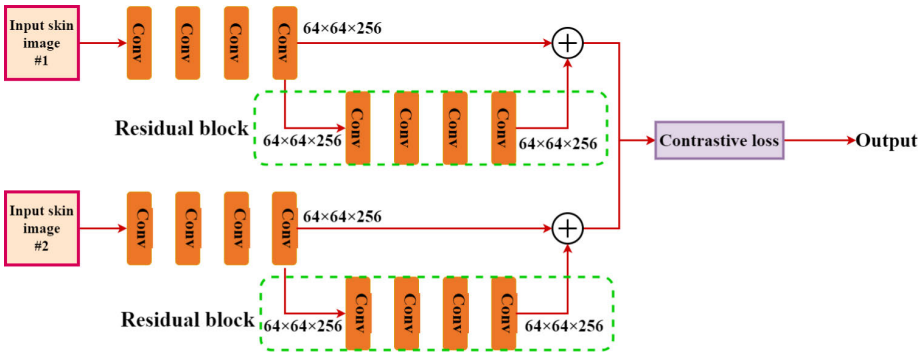


FIGURE 5. An overview of the proposed method on skin images from the perspective of convolutional layers. Each of the convolutional layers in the figure is named *Conv*.

TABLE 2. Comparison between the cutting-edge methods and Breast-twins model. This comparison is on BreaKHis at 400× magnification.

Method	K	Accuracy	Precision	F1-score
Breast-twins	1	0.59	0.67	0.70
Breast-twins	3	0.83	0.81	0.88
Breast-twins	5	0.92	0.90	**0.94**
MCCH [33]	5	-	0.89	-
FedCBMIR [4]	5	0.96	**0.94**	-
CNN based CAE [33]	5	-	0.93	-
UCBMIR [24]	5	-	0.91	-
VTHC [34]	6	**0.99**	-	-
LBSE [35]	9	-	0.77	-
HSDH [20]	400	-	**0.99**	-

TABLE 3. Quantitative comparison between the proposed Breast-twins model and the baseline CAE at *K*=1 and *K*=3, in terms of Accuracy, Recall, Precision, and F1-score. The results demonstrate the improved retrieval performance of the Siamese-based Breast-twins architecture, particularly at Top-1 evaluation.

Method	K	Accuracy	Recall	Precision	F1-score
CAE	1	0.48	0.54	0.64	0.56
Breast-twins	1	**0.58**	**0.73**	**0.67**	**0.70**
CAE	3	0.83	0.95	**0.82**	0.88
Breast-twins	3	**0.83**	**0.97**	0.81	**0.88**

this as a limitation and an important direction for future work in collaboration with our clinical partners.

A. RESULTS OF BREAST-TWINS

Table 2 and Table 3 provide a comparison between some state-of-the-art papers that reported the results with different amounts of K ($K = 1, 3, 5$). In these tables, the value of K for each study is specified, as this parameter has not been extensively examined in previous works, which typically report results only for the top-1 retrieval on breast datasets. As shown in Table 2, the Breast-twins model demonstrates strong retrieval performance on the BreaKHis dataset at 400× magnification, achieving a Top-1 accuracy of 59%. This indicates that in 59% of cases, the most similar image retrieved was from the correct class. Notably, the model achieves 83% accuracy at Top-3 and 92% at Top-5, reflecting its robustness and reliability across different K.

When comparing with existing methods, Breast-twins surpasses MCCH [33] at $K = 5$ in terms of precision, achieving 90% versus 89%. Furthermore, the Breast-twins model performs competitively with HSDH [20], which uses a considerably higher amount of K ($K = 400$), reporting a precision of 99%. Despite this, the precision of Breast-twins at $K = 5$ is only 0.09 lower, showing that Breast-twins achieves comparable results with a much smaller and more practical retrieval set—important for clinical scenarios where reviewing hundreds of results is not feasible.

Another noteworthy comparison is with FedCBMIR [4], a model trained on all four magnification levels of BreaKHis, giving it a broader generalization capability. Even so, when tested specifically at 400×, FedCBMIR outperforms Breast-twins by only 0.04 in Top-5 accuracy (96% vs. 92%). This small gap highlights the effectiveness of Breast-twins, especially considering it was trained exclusively on a single magnification level.

VTHSC-MIR, another recent method, achieves a high accuracy of 99% but uses a Top-6 retrieval setting. While this may seem advantageous, it shifts the focus to larger

TABLE 4. Comparison between the cutting-edge methods and Breast-twins model. In this comparison, Breast-twins is on BreaKHis at 400× magnification, and all others are reported on all four magnifications of the BreaKHis data set. $K = 10$.

Method	Precision
Local binary pattern [16]	0.50
Pre-trained CNN [16]	0.54
Semantic patient scores [16]	0.61
Trained CNN [36]	0.62
Semantic class scores [16]	0.80
Breast-twins	**0.95**

retrieval pools. In contrast, our Siamese-based CBHIR model emphasizes performance at lower values of K—specifically Top-1 and Top-3—which are more clinically relevant, as real-time decision support systems often rely on the most immediate matches. Retrieving relevant results in the top few positions is significantly more challenging and critical in practical use cases.

Additionally, the lightweight and targeted architecture of the Breast-twins model helps prevent overfitting, a common risk when working with small and imbalanced medical datasets like BreaKHis. This ensures our model remains efficient and generalizable, even without massive training data or complex ensembles.

As previously mentioned, no prior studies have reported performance at Top-1 retrieval, which posed a challenge when comparing our results at this level. To address this, we trained and tested a CAE model identical to the one proposed in [24], using the same configurations and protocols. This allowed us to directly compare its Top-1 retrieval performance with that of our models. For clarity, we refer to this retrained model as ''CAE'' in our tables. The baseline CAE model was reproduced using the same configuration, hyperparameters, and training procedures described in our previous publication [24], ensuring a fair and unbiased comparison. Since the original CAE framework was developed by our team, we utilized the same codebase and dataset split, modifying only the evaluation to K=1 for a consistent Top-1 comparison with the proposed method.

The corresponding results are presented in Table 3. Although the CAE demonstrates competitive accuracy, precision, and F1-score at Top-3, its performance significantly drops at Top-1 retrieval. This highlights a critical limitation: CBHIR models optimized only for high K values may fail to deliver precise results in immediate retrievals—potentially reducing their clinical utility. Based on this table, the Breast-twins model balances precision, practicality, and computational efficiency, making it a promising candidate for real-world histopathological image retrieval systems.

Table 4 provides a comparison at $K = 10$ with some state-of-the-art techniques, which made the search and reported the results at all its magnifications of BreaKHis. As can be understood from Table 4, Breast-twins could surpass them even if training on BreaKHis at all magnifications offers a more generalized model.

Figure 6 demonstrates the superiority of Breast-twins over others (e.g., KNN) in terms of retrieval precision. In this figure, each method has a corresponding data point, labeled with its precision value and depicted in different colors. Among the methods, Breast-twins has the highest precision at 0.9, shown in red, while the lowest precision is for KNN at 0.758, shown in salmon. We named Method1 and Method2 in Figure 6 because the authors in [37] did not identify their two approaches. We next contrasted their outcomes with the top 20 retrieved images and our results with the top 5. Breast-twins model was able to surpass the other approaches and had superior precision in the retrieval performance. It is noteworthy to notice that Methods 1 and 2 in [37] are supervised hashing techniques that achieved 87% and 89.5% precision by recommending a multi-feature supervised hashing method. The methods in [37] were reported at top 20 retrieval.

Finding a sufficient amount of K is challenging for DL experts. For instance, in the case of [20], the results were reported at the top 400, which delivered 99% accuracy for a binary data set. According to the definition of accuracy in CBHIR, increasing the amount of K yields higher accuracy. But a high amount of K, means retrieving images with longer distances and less similarity. Therefore, a CBHIR with high accuracy at a high amount of K cannot provide a confident tool. The impacts of the amount of K can be seen in Figure 7 which shows three confusion matrices at top 1, 3, and 5 for breast cancer. The trends in these confusion matrices highlight the trade-offs between precision and recall in CBHIR which means a higher value of K increases recall, but it might decrease precision. While recall means retrieving more relevant images and precision means results including more non-relevant images. So, if the amount of K is small, it balances the retrieval of true positives and false positives. However, when K increases, precision is challenged, resulting in fewer true negatives and potentially more false positives, which aligns with the behavior of the confusion matrices.

Figure 8 plots the T-distributed Stochastic Neighbor Embedding (TSNE) of the extracted features of the breast data set. TSNE is a data visualization technique that is used for reducing the dimensionality of high-dimensional data while preserving the similarities between data points. Figure 8 compares the extracted embedding space as a result of FedCAE [4] and the proposed Breast-twins. FedCAE was trained on all four magnifications of the BreaKHis data set and then applied to BreaKHis at 400× magnification. As can be seen in Figure 8, although FedCAE is more generalized due to the different magnifications, the embedding space resulting from Breast-twins is more discriminative.

In breast cancer diagnosis, pathologists need to take into account the histopathological patterns of the tissue to be able to grade the tissue. Well-differentiated patterns, low nuclear pleomorphism, and mitotic activities are some of the important criteria that pathologists analyze in their query to write a diagnosis report [6]. These grading patterns

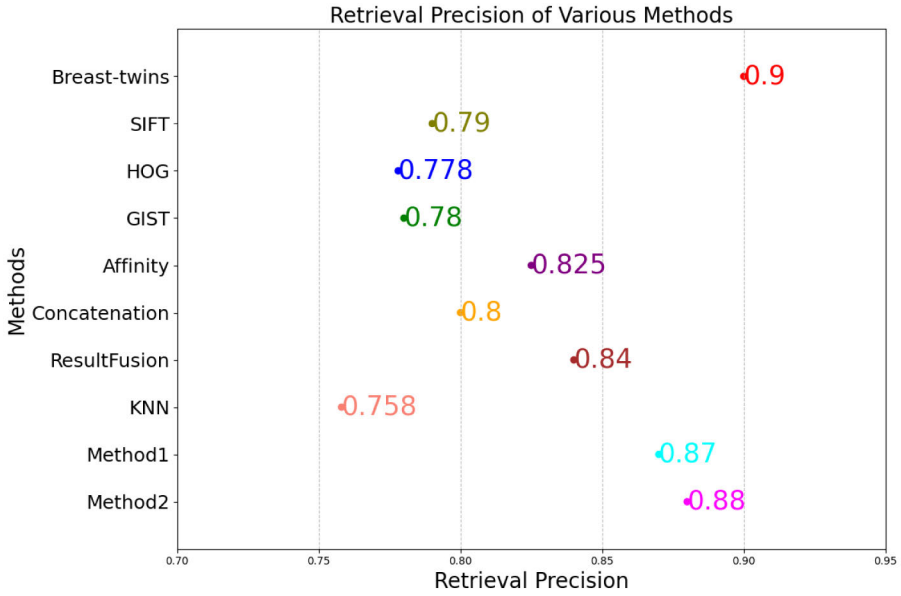


FIGURE 6. An indirect comparison between the results of Breast-twins and some recent methods for different amounts of K. Each point in this figure illustrates the retrieval precision obtained by various methods, including KNN, SIFT, and others.

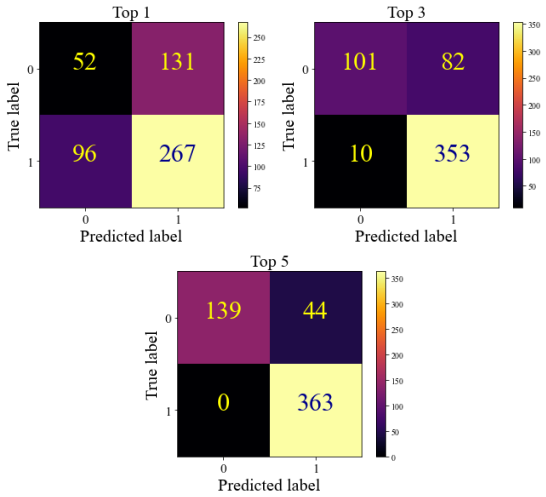
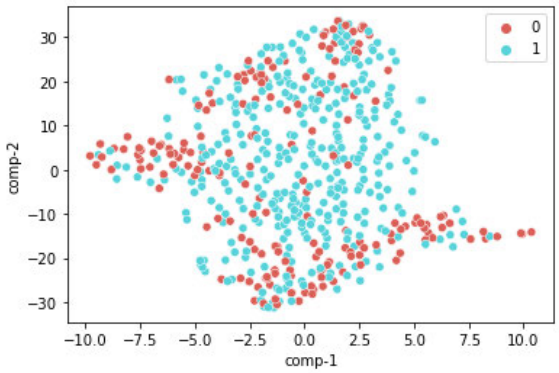
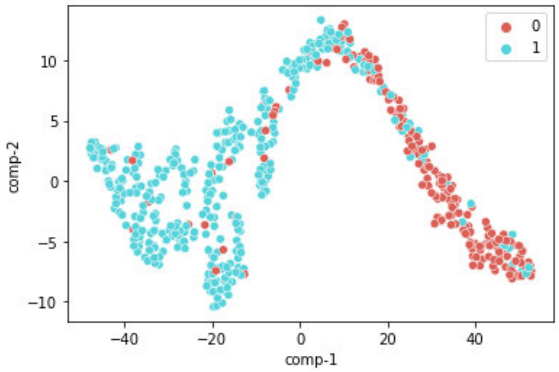


FIGURE 7. Three confusion matrices at top 1, 3, and 5 for the Breast-twins CBHIR model.

assist pathologists in determining the best treatment and predicting the likelihood of the tumor spreading. So, the CBHIR system answers these demands from pathologists by providing similar patterns for the pathologists. Figure 9 shows eight random patches of the BreaKHis data set with their Grad-CAM plots. The presented Grad-CAM visualizations were generated by fusing the GMP layer of the proposed Breast-twins network to emphasize the final feature representation of the model. It is important to note that the Grad-CAM visualizations presented in this study are qualitative and intended to illustrate the regions emphasized by the Siamese embeddings during image retrieval, rather than to provide a quantitative measure of interpretability.



(a) Embedding space extracted by FedCBMIR



(b) Embedding space extracted by Breast-twins

FIGURE 8. TSNE plot of the extracted feature space by applying two different FE to the breast data set. Zero label in red shows the embedding space of benign tissues and the One labeled points in blue corresponds to the malignant cases.

Since pathologist-annotated regions of interest were not available, quantitative validation, such as calculating the

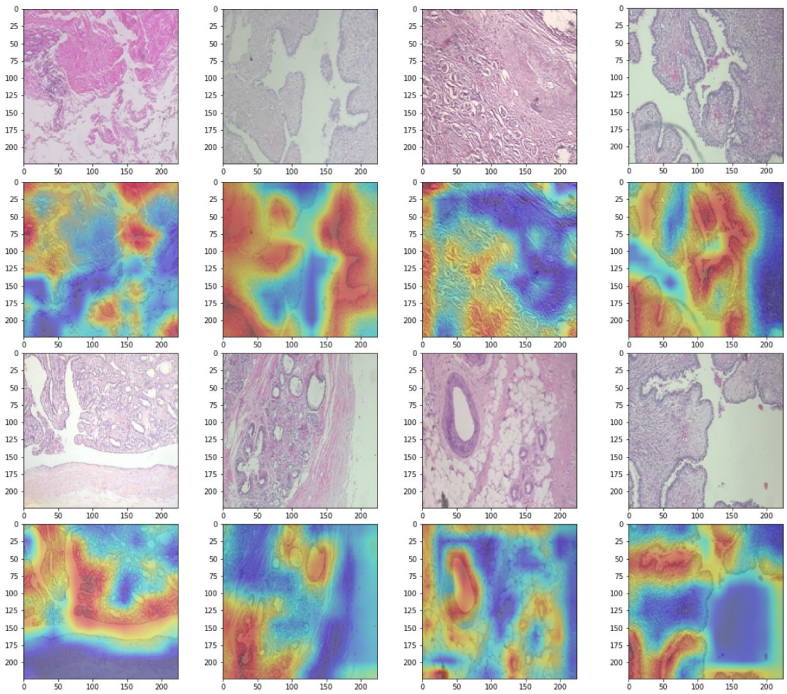


FIGURE 9. Eight random patches of breast cancer data set. The odd rows show the original images. The even rows are the Grad-CAMs mapped on top of the original images.

Intersection over Union (IoU) between Grad-CAM maps and expert annotations, remains an important direction for future work.

These Grad-CAM plots provide insight into the DL-based Siamese networks to clarify why the model made a particular decision. This means that it can identify if the model focused on the correct histopathological features or retrieved the images based on the artifact. Figure 9 clarifies the important regions of the tissue that the Breast-twins network detected as an important part of the patches for the final decision of the search engine in the CBHIR model. This figure proves the interpretability of the proposed model.

Figure 10 visualizes the output of the proposed CBHIR by using Breast-twins for three random queries. The retrieved images were compared with the query based on their labels. Images with a label different than the query are marked in red. It is seen that the proposed Breast-twins network can return the patches with similar histopathological features, with the correct class labels of these patches. Results indicate that the Breast-twins network is highly efficient in retrieving similar patches.

B. RESULTS OF SKIN-TWINS

Table 5 presents the evaluation of the Skin-twins model in retrieving similar patches among malignant and benign skin lesions at the top 3 and top 5 retrieved images. As observed, the Skin-twins achieves an F1-score of 0.94 and an overall accuracy of 89% at Top-3 retrieval. The F1-score further improves to 0.96 and the accuracy to 93% at Top-5. This trend confirms that the model consistently retrieves highly relevant images, with precision and recall increasing as the number of considered retrievals (K) increases. It is important to highlight that these high F1-scores at relatively low K values demonstrate the model's strong capability to discriminate between benign and malignant cases with minimal retrieval burden, offering immediate and reliable second opinions for clinical decision-making. Moreover, the relatively small gap between Top-3 and Top-5 results suggests that the model maintains robustness and does not significantly degrade in retrieval quality even as K increases slightly. The results from 5 demonstrate the high retrieval accuracy of the Skin-twins model across different values of K, indicating its robustness and practical efficiency for clinical applications. To further validate the strength of the proposed approach, Table 6 provides a direct comparison at the most challenging retrieval level, Top-1. Unlike many prior works that rely on larger K values to report high retrieval scores, our Skin-twins model shows substantial superiority even when evaluated at the first retrieved image, where no margin for error exists.

Table 6 compares the Skin-twins CBHIR model against a previously proposed Convolutional Autoencoder (CAE)-based CBHIR method for skin cancer diagnosis at Top-1 retrieval ($K = 1$). The Skin-twins framework outperforms the CAE-based method by a wide margin across all evaluation metrics. Notably, the Skin-twins model achieves a precision of 80% compared to 0.13 for the CAE model — an improvement of over 6 times. Similarly, recall and F1-score values are significantly higher in the Skin-twins model,

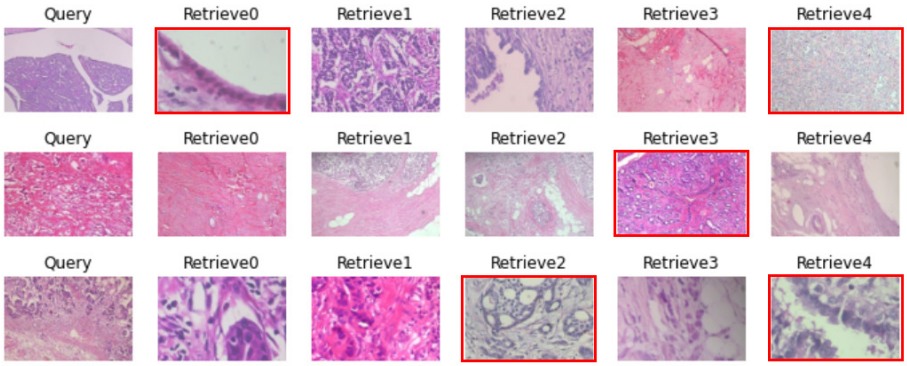


FIGURE 10. Three random breast queries with their 5 top retrieved images. The miss-retrieved images are marked in red.

TABLE 5. Results of Skin-twins in searching between "*Malignant*" and "*Benign*" at top 3 and 5. ($K = 3, 5$).

Method	K	Precision	Recall	F1-score		F1s-avg	Accuracy
Skin-twins	3	0.88	0.99	0.64	0.94	0.94	0.89
	5	0.92	1	0.80	0.96	0.96	0.93

TABLE 6. A comparison between the results of Skin-twins and state-of-the-art studies at the first top retrieval ($K = 1$).

Method	Precision	Recall	F1-score	Accuracy
CAE	0.13	0.14	0.135	**0.72**
Skin-twins	**0.80**	**0.82**	**0.81**	0.69

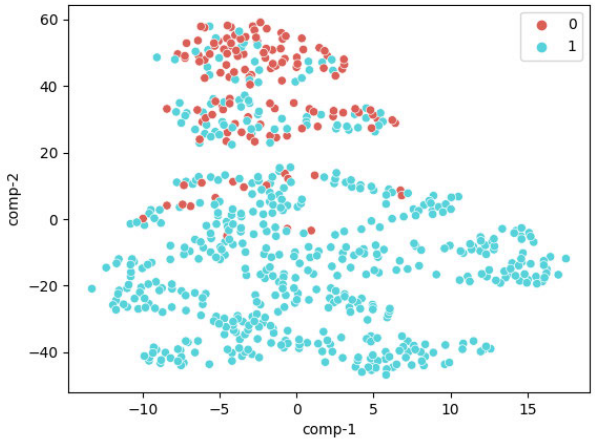


FIGURE 11. TSNE plot of the extracted features of the skin data set by Skin-twins network. Features corresponding to benign tissues are plotted in red and labeled as **Zero**. The blue points are related to the malignant tissues, which are labeled as **One**.

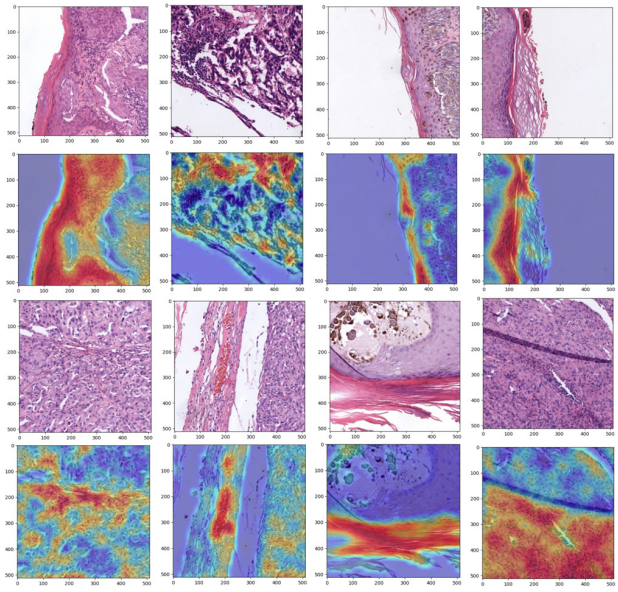


FIGURE 12. Eight random patches of skin cancer data set. The odd rows show the original images. The even rows are the Grad-CAMs mapped on top of the original images.

indicating its superior ability to retrieve relevant patches with high confidence at the first attempt. This direct comparison at Top-1 retrieval is particularly crucial because it reflects the model's ability to retrieve the correct histopathological match without relying on larger retrieval pools, which is critical for practical clinical use where pathologists require immediate, high-precision feedback.

Figure 11 shows the TSNE plot of the extracted features of the skin data set. Each point in the plot represents an individual image from the skin cancer data set, mapped to the feature space into a two-dimensional plane for better interpretability. Red points represent benign samples (labeled as 0), while blue points correspond to malignant samples (labeled as 1). As seen in the figure, the Skin-twins network is able to create a well-separated embedding space, where benign and malignant samples form distinct clusters with minimal overlap. This clear separation demonstrates that the proposed network has successfully learned a discriminative

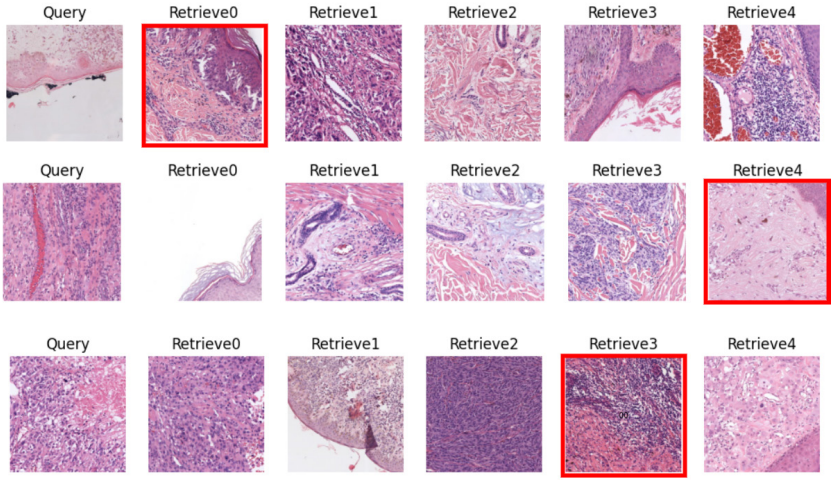


FIGURE 13. Three random images with their top 5 retrieved images. The returned images with a different label are marked in red.

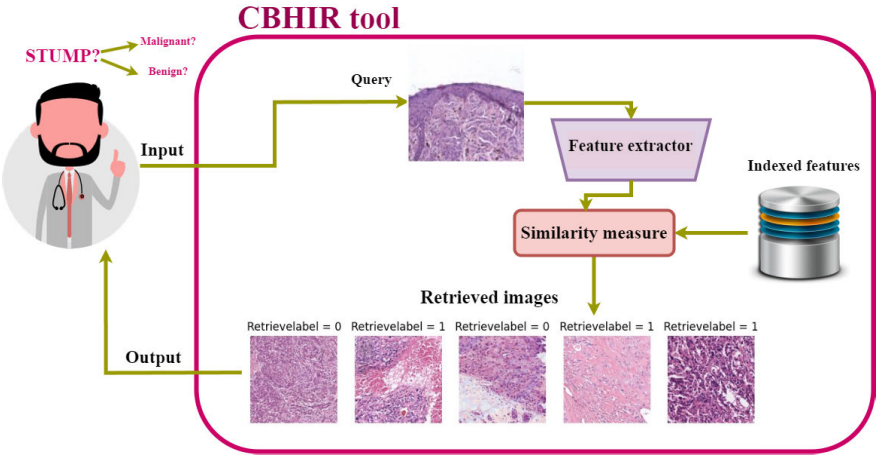


FIGURE 14. A real-world scenario that a pathologist faces with a STUMP case and uploads the query as input to the proposed CBHIR approach. First, the model feeds the query to the proposed feature extractor and extracts the query features. Second, a similarity measure applies to the query features and indexed features of the database. Then, it is time to rank and visualize the top 5 similar patches to pathologists as the output of the proposed CBHIR.

feature representation capable of capturing the underlying histopathological differences between benign and malignant skin tissues. Such structured embedding is critical for effective retrieval tasks because it ensures that similar images (i.e., within the same class) are mapped closer together, while dissimilar images (from different classes) are mapped farther apart. The well-formed clusters in the TSNE plot further support the quantitative results shown earlier, confirming that the Skin-twins model not only achieves high retrieval performance but also organizes the feature space in a clinically meaningful and interpretable way.

The Grad-CAM plots of the Skin-twins are shown in Figure 12. In skin cancer detection, Cellularity, Nuclear Features, Cellular Morphology, Architecture, Mitotic activity, Stroma and Connective Tissue, etc., are some of the histological features that pathologists consider to grade a skin cancer [38]. These are the histological patterns that the CBHIR should focus on to find similar patches. Figure 12 shows the success of the model in detecting the cancerous and abnormal regions of the tissue. For instance, in some cases, the main focus is cellularity and assessing the density of the cells within the tissue. High cellularity may indicate increased cell proliferation or inflammation. So, the Grad-CAM plots generate a visually interpretable explanation for the retrieved images, which explains why the Skin-twins model retrieved the images for the pathologists.

Figure 13 depicts three query examples of the skin data set with their retrieved images. The red line around the retrieved patches clarifies that the retrieved image does not have the same label as the query. As can be seen in this figure, just one out of five retrieved patches was not at the same cancer grade as the query.

C. STUMP SEARCHING

Spitzoid lesions present a diagnostic conundrum due to their intricate histology, creating challenges in establishing clear

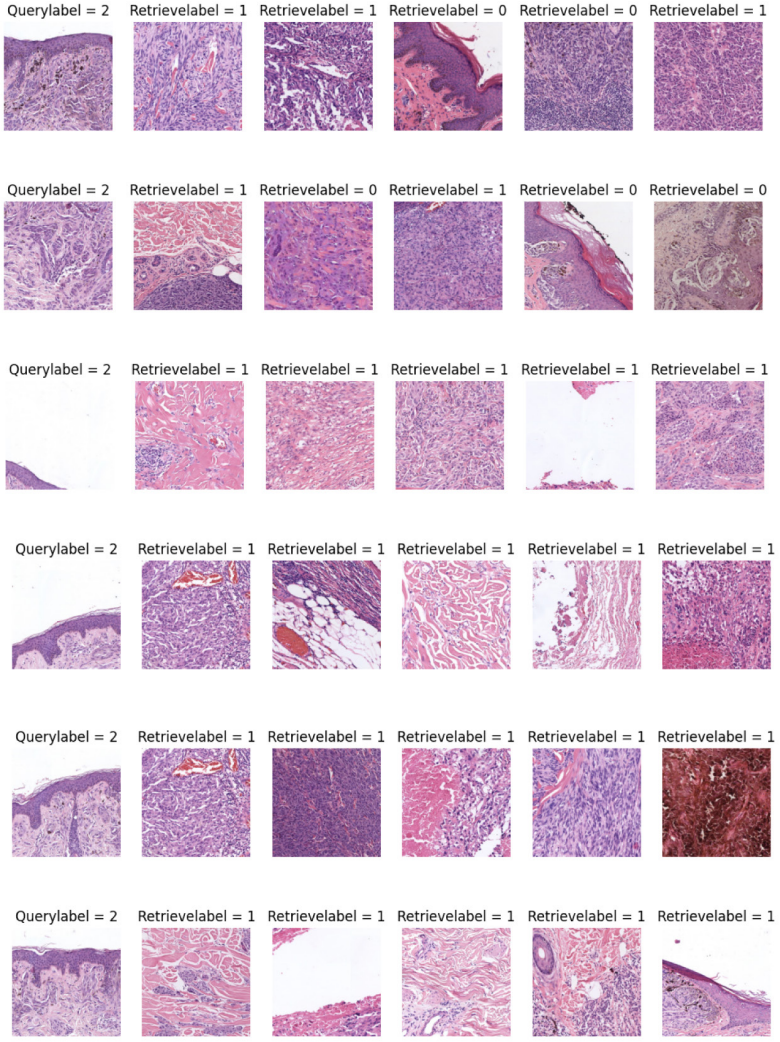


FIGURE 15. Six random examples of STUMP queries and their retrieved images. The STUMP queries were labeled = 2, Malignant = 1, and Benign = 0.

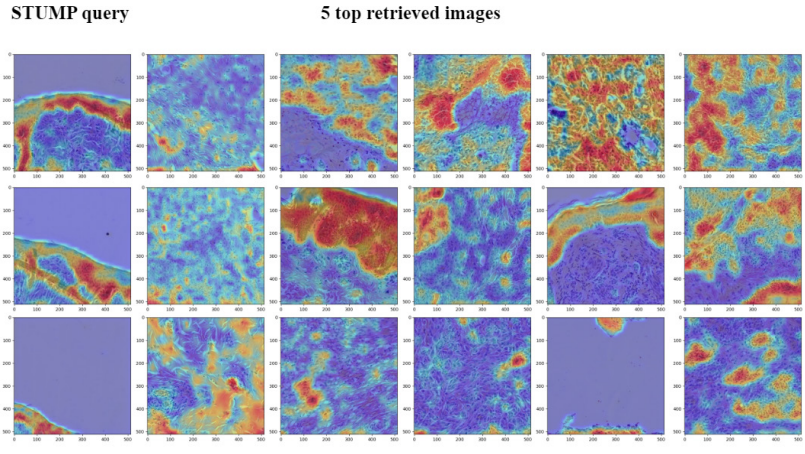


FIGURE 16. Three random STUMP patches with their top 5 retrieved image. The Grad-CAMs show the main patterns that the CBHIR took into account to find the similarity between the query and the retrieved patches.

parameters between benign nevi and potentially malignant melanomas. In particular, STUMP cases require careful assessment to determine their true nature. In this paper, the model was isolated from STUMP images during the

training and testing steps for the above experiments. Figure 14 explains that this experiment mimics the situation that pathologists face with a STUMP tissue and need a second opinion to assist them in writing the diagnosis report. To do so, the Skin-twins model, which is trained on the binary data set including *Benign* and *Malignant*, is fed by the STUMP queries. The Skin-twins model displays top K similar patches to pathologists. The top K return patches with their labels, providing some clues and hints regarding histopathological features and patterns for the pathologists.

Due to the limited number of STUMP cases, we did not perform a formal statistical analysis. However, as shown in Figure 15, this tool can provide useful clinical guidance to check the simialr pathological patterns of the previous cases. The main goal of including this experiment and sharing its results is to demonstrate a real-world use case of the proposed tool and show how it can assist pathologists in addressing the grading challenges. This example highlights the practical value of our framework as a transparent and interpretable tool, offering meaningful visual and retrieval-based insights, rather than relying on a black-box classifier. In this way, the proposed system can support pathologists in making more informed and explainable diagnostic decisions.

In Figure 15, for each query STUMP (labeled = 2), five top similar patches and their labels are retrieved from the data set, including *Benign* (labeled = 0) and *Malignant* (labeled = 1). In this case, pathologists can analyze the patterns in the retrieved tissues and compare them with the histopathological features of their STUMP query. The final decision depends on the pathologists' point of view and their knowledge. For instance, in the case of line 3 in Figure 15, it can be concluded that the STUMP tissue is Malignant since all the retrieved images are labeled = 1.

In Figure 16, there are Grad-CAM plots of three random STUMP queries with their top 5 retrieved images. The histopathological patterns and the regions of interest at the patch level that the Skin-twins network paid attention to retrieve the patches related to the STUMP queries are highlighted.

Figure 17 shows the confusion matrices of the proposed method on the skin data set at top $K = 1, 3$, and 5. In these confusion matrices, *Benign* and *Malignant* are labeled as zero and one. The figure illustrates that by increasing the amount of K, the amounts of True positive and True negative are increasing. This consistent performance across both Top-3/5 and Top-1 levels highlights the reliability and effectiveness of the proposed Siamese-based CBHIR system for real-world diagnostic support in dermatopathology.

D. COMPARISON WITH CLASSIFICATION APPROACHES

Classification delivers the label of the input image to provide a second opinion on the query tissue for the pathologists. While CBHIR not only provides the label but also retrieves similar patches. The outputs of CBHIR assist pathologists in finding the relevant histopathological features in the returned images.

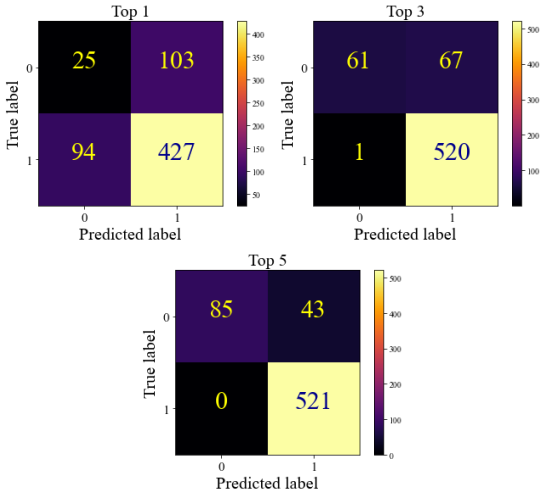


FIGURE 17. Three confusion matrices at top 1, 3, and 5 for Skin-twins CBHIR model.

The only similarity between a CBHIR and a classifier is that both report the labels. A classifier has done this as a black box, while in CBHIR, pathologists have the opportunity to analyze the retrieved patches by their knowledge to understand why the model retrieves images. So, a classification framework, by its nature, directly reports the label. However, in CBHIR, the label is obtained based on the label of the retrievals and the final decision of the pathologists by analyzing the patterns of the query and the returned images from a CBHIR model.

In order to evaluate the performance of the CBHIR, the retrieved images have to be in the same cancer type as the query. Table 7 presents a comparative analysis between the Skin-twins CBHIR model and the classifiers reported in [32] which have been recently published.

In this table, the results of Skin-twins are reported in the top 3 retrieved images. The obtained results of Skin-twins CBHIR could exhibit higher performance compared with all the mentioned methods in terms of F1-score. Furthermore, assume that K was set as 5 as the most used amount of K in the previous studies [4], [17], [20], [22], [23], [24], [25], the F1-score and accuracy are 96% and 93%, respectively, much higher than the results of classifiers.

Table 8 provides the comparison between the Breast-twins CBHIR model at the top 5 and the recent classifiers on the BreaKHis data set. Breast-twins CBHIR could reach a higher accuracy than the recent classifiers, although, in some studies, it is slightly under-performing a classifier.

For instance, in [39], the authors trained the model in two stages: first as an autoencoder (AE) and then re-trained it as a Siamese classifier. This dual-stage training pipeline increased computational cost and GPU usage, yet only resulted in a marginal improvement of approximately 0.04 in accuracy compared to our proposed Breast-twins CBHIR model. Compared to Liu et al., who rely on pretraining and multi-scale Gaussian pyramids for feature extraction, our approach uses a single, custom Siamese network trained end-to-end on fixed high magnification images (400×), providing

TABLE 7. A comparison between classifiers and the Skin-twins CBHIR, $K = 3$.

Methods	F1-score	Accuracy
VGG16 [32]	0.70	0.72
ResNet-34 [32]	0.84	0.85
resNet-50 [32]	0.76	0.77
Resnet-34 ($\overset{\alpha}{S}$) [32]	0.74	0.68
Skin-twins	**0.94**	**0.90**

TABLE 8. A comparison between classifiers and the Breast-twins CBHIR, $K = 3$.

Methods	Accuracy
KSH [35]	0.48
R2SDH [35]	0.50
SCDH [35]	0.69
SDHMLR [35]	0.69
LBSE [35]	0.80
PFTAS + SVM [40]	0.823
IDSNet [41]	0.845
LBSE$_F$ [35]	0.85
Breast-twins	0.92
FE-VGGNET16-SVM(POLY) [42]	0.934
FCN-Bi-LSTM [43]	0.942
AE + Siamese Network [39]	0.967

a more consistent and efficient feature representation. While their method is classification-oriented, our model is designed for content-based image retrieval, leveraging contrastive loss to better capture inter-image similarity, with an emphasis on real-time clinical applicability. Additionally, our models—Breast-twins and Skin-twins—are capable of retrieving images strictly within the same cancer type, as demonstrated in Tables 8 and 7, further supporting the reliability and specificity of our approach. As can be seen in these tables, our method offers a more practical, resource-efficient, and retrieval-oriented solution for histopathological image analysis.

V. CONCLUSION AND FUTURE LINES

In this paper, we proposed a novel approach to Content-Based Histopathological Image Retrieval (CBHIR) for breast and skin cancer, using a custom-designed Siamese network framework called Breast-twins and Skin-twins. This approach focuses on retrieving similar patches in histopathological images to support pathologists in diagnosis. By employing contrastive loss, the Siamese network effectively extracts distinguishing features that aid in image retrieval, ranking patches based on Euclidean distance from a query image. Our results, evaluated at top-1, top-3, and top-5 retrieved patches, show that our model outperforms other techniques, especially at the top-1 level, providing reliable performance in retrieving similar images for breast and skin cancer.

For breast cancer, the Breast-twins model was trained using the BreaKHis dataset at 400× magnification, where it supports mitosis evaluation—an essential factor in cancer grading. The model demonstrated significantly improved retrieval performance compared to existing CBHIR methods. Specifically, Breast-twins achieved a 70% F1-score at Top-1 retrieval, outperforming recent approaches such as the Convolutional Autoencoder (CAE), which reported a 56% F1-score. Moreover, at Top-10 retrieval, Breast-twins attained a 95% precision, reflecting a 15% improvement over the state-of-the-art at the same number of retrieved images.

Similarly, for skin cancer, Skin-twins substantially outperformed the Convolutional Autoencoder (CAE)-based method across multiple metrics, including precision, recall, and F1-score. Notably, Skin-twins achieved a precision of 80%, representing a 67% improvement over the CAE model. These results underscore the effectiveness of our approach in handling diagnostically challenging cases such as Spitzoid Tumors of Uncertain Malignant Potential (STUMP), where retrieving top-K similar patches can assist pathologists in making faster and more reliable evaluations.

Our findings demonstrate that CBHIR models like Breast-twins and Skin-twins hold great potential to support histopathological diagnosis in clinical workflows. Future work could explore expanding data diversity through federated learning, creating a multi-magnification retrieval model for breast cancer, and integrating text-based data. These advancements can further refine CBHIR systems, enabling broader and more precise applications in digital pathology.

VI. AUTHOR CONTRIBUTIONS

Zahra Tabatabaei: Conceptualization, Methodology, Analyzing, Writing–Original draft, Reviewing and editing, Formal analysis, visualization.

Adrián Colomer: Supervision and Review.

Javier Oliver Moll: Supervision.

Valery Naranjo: Supervision.

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REFERENCES

[1] Z. Apalla, A. Lallas, E. Sotiriou, E. Lazaridou, and D. Ioannides, "Epidemiological trends in skin cancer," *Dermatol. Practical Conceptual*, vol. 7, no. 2, pp. 1–6, Apr. 2017.

[2] J. Reichrath, U. Leiter, T. Eigentler, and C. Garbe, "Epidemiology of skin cancer," in *Sunlight, Vitamin D and Skin Cancer*. New York, NY, USA: Springer, 2014, pp. 120–140.

[3] T. Wiesner, H. Kutzner, L. Cerroni, M. C. Mihm, K. J. Busam, and R. Murali, "Genomic aberrations in spitzoid melanocytic tumours and their implications for diagnosis, prognosis and therapy," *Pathology*, vol. 48, no. 2, pp. 113–131, Feb. 2016.

[4] Z. Tabatabaei, Y. Wang, A. Colomer, J. Oliver Moll, Z. Zhao, and V. Naranjo, "WWFedCBMIR: World-wide federated content-based medical image retrieval," *Bioengineering*, vol. 10, no. 10, p. 1144, Sep. 2023.

[5] R. Rasti, M. Teshnehlab, and S. L. Phung, "Breast cancer diagnosis in DCE-MRI using mixture ensemble of convolutional neural networks," *Pattern Recognit.*, vol. 72, pp. 381–390, Dec. 2017.

[6] Y. Ma, Z. Jiang, H. Zhang, F. Xie, Y. Zheng, H. Shi, and Y. Zhao, "Breast histopathological image retrieval based on latent Dirichlet allocation," *IEEE J. Biomed. Health Informat.*, vol. 21, no. 4, pp. 1114–1123, Jul. 2017.

[7] N. Kanwal, T. Eftestøl, F. Khoraminia, T. C. Zuiverloon, and K. Engan, "Vision transformers for small histological datasets learned through knowledge distillation," in *Proc. Pacific–Asia Conf. Knowl. Discovery Data Mining*. Cham, Switzerland: Springer, 2023, pp. 167–179.

[8] H. R. Shahdoosti and Z. Tabatabaei, "Object-based feature extraction for hyperspectral data using firefly algorithm," *Int. J. Mach. Learn. Cybern.*, vol. 11, no. 6, pp. 1277–1291, Jun. 2020.

[9] H. R. Shahdoosti and Z. Tabatabaei, "MRI and PET/SPECT image fusion at feature level using ant colony based segmentation," *Biomed. Signal Process. Control*, vol. 47, pp. 63–74, Jan. 2019.

[10] A. Ahmed, "Implementing relevance feedback for content-based medical image retrieval," *IEEE Access*, vol. 8, pp. 79969–79976, 2020.

[11] Z. Tabatabaei, A. Colomer, K. Engan, J. Oliver, and V. Naranjo, "Self-supervised learning of a tailored convolutional auto encoder for histopathological prostate grading," in *Proc. 31st Eur. Signal Process. Conf. (EUSIPCO)*, Sep. 2023, pp. 980–984, doi: 10.23919/eusipco58844.2023.10289741.

[12] H. Lee and Y.-P.-P. Chen, "Image based computer aided diagnosis system for cancer detection," *Expert Syst. Appl.*, vol. 42, no. 12, pp. 5356–5365, Jul. 2015, doi: 10.1016/j.eswa.2015.02.005.

[13] X. Li, J. Yang, and J. Ma, "Recent developments of content-based image retrieval (CBIR)," *Neurocomputing*, vol. 452, pp. 675–689, Sep. 2021, doi: 10.1016/j.neucom.2020.07.139.

[14] Y. Xing, B. J. Meyer, M. Harandi, T. Drummond, and Z. Ge, "Multimorbidity content-based medical image retrieval and disease recognition using multi-label proxy metric learning," *IEEE Access*, vol. 11, pp. 50165–50179, 2023.

[15] X. Zhang, K. Liu, A. S. Kumar, and M. Rahman, "A study of the diamond tool wear suppression mechanism in vibration-assisted machining of steel," *J. Mater. Process. Technol.*, vol. 214, no. 2, pp. 496–506, Feb. 2014.

[16] E. Mahraban Nejad, L. S. Affendey, R. B. Latip, I. B. Ishak, and R. Banaeeyan, "Transferred semantic scores for scalable retrieval of histopathological breast cancer images," *Int. J. Multimedia Inf. Retr.*, vol. 7, no. 4, pp. 241–249, Nov. 2018.

[17] Z. Tabatabaei, A. Colomer, K. Engan, J. Oliver, and V. Naranjo, "Residual block convolutional auto encoder in content-based medical image retrieval," in *Proc. IEEE 14th Image, Video, Multidimensional Signal Process. Workshop (IVMSP)*, Jun. 2022, pp. 1–5.

[18] J. C. Caicedo, F. A. González, and E. Romero, "Content-based histopathology image retrieval using a kernel-based semantic annotation framework," *J. Biomed. Informat.*, vol. 44, no. 4, pp. 519–528, Aug. 2011.

[19] X. Zhang, W. Liu, M. Dundar, S. Badve, and S. Zhang, "Towards large-scale histopathological image analysis: hashing-based image retrieval," *IEEE Trans. Med. Imag.*, vol. 34, no. 2, pp. 496–506, Feb. 2015.

[20] S. M. Alizadeh, M. S. Helfroush, and H. Müller, "A novel Siamese deep hashing model for histopathology image retrieval," *Expert Syst. Appl.*, vol. 225, Sep. 2023, Art. no. 120169.

[21] X. Wang, Y. Du, S. Yang, J. Zhang, M. Wang, J. Zhang, W. Yang, J. Huang, and X. Han, "RetCCL: Clustering-guided contrastive learning for whole-slide image retrieval," *Med. Image Anal.*, vol. 83, Jan. 2023, Art. no. 102645.

[22] Y. Zheng, Z. Jiang, H. Zhang, F. Xie, Y. Ma, H. Shi, and Y. Zhao, "Size-scalable content-based histopathological image retrieval from database that consists of WSIs," *IEEE J. Biomed. Health Informat.*, vol. 22, no. 4, pp. 1278–1287, Jul. 2018.

[23] N. Hegde, J. D. Hipp, Y. Liu, M. Emmert-Buck, E. Reif, D. Smilkov, M. Terry, C. J. Cai, M. B. Amin, C. H. Mermel, P. Q. Nelson, L. H. Peng, G. S. Corrado, and M. C. Stumpe, "Similar image search for histopathology: SMILY," *npj Digit. Med.*, vol. 2, no. 1, p. 56, Jun. 2019.

[24] Z. Tabatabaei, A. Colomer, J. O. Moll, and V. Naranjo, "Toward more transparent and accurate cancer diagnosis with an unsupervised cae approach," *IEEE Access*, vol. 11, pp. 143387–143401, 2023.

[25] S. Kalra, H. R. Tizhoosh, C. Choi, S. Shah, P. Diamandis, C. J. V. Campbell, and L. Pantanowitz, "Yottixel—An image search engine for large archives of histopathology whole slide images," *Med. Image Anal.*, vol. 65, Oct. 2020, Art. no. 101757.

[26] S. Hemati, S. Kalra, M. Babaie, and H. R. Tizhoosh, "Learning binary and sparse permutation-invariant representations for fast and memory efficient whole slide image search," *Comput. Biol. Med.*, vol. 162, Aug. 2023, Art. no. 107026.

[27] N. Kayhan and S. Fekri-Ershad, "Content based image retrieval based on weighted fusion of texture and color features derived from modified local binary patterns and local neighborhood difference patterns," *Multimedia Tools Appl.*, vol. 80, nos. 21–23, pp. 32763–32790, Sep. 2021.

[28] M. Shorfuzzaman and M. S. Hossain, "MetaCOVID: A Siamese neural network framework with contrastive loss for n-shot diagnosis of COVID-19 patients," *Pattern Recognit.*, vol. 113, May 2021, Art. no. 107700.

[29] F. Meng, X. Gong, and Y. Zhang, "SiamRank: A Siamese based visual tracking network with ranking strategy," *Pattern Recognit.*, vol. 141, Sep. 2023, Art. no. 109630.

[30] R. Singh, V. Bharti, V. Purohit, A. Kumar, A. K. Singh, and S. K. Singh, "MetaMed: Few-shot medical image classification using gradient-based meta-learning," *Pattern Recognit.*, vol. 120, Dec. 2021, Art. no. 108111.

[31] I. Pacal and O. Attallah, "Hybrid deep learning model for automated colorectal cancer detection using local and global feature extraction," *Knowledge-Based Syst.*, vol. 319, Jun. 2025, Art. no. 113625.

[32] L. Launet, A. Colomer, A. Mosquera-Zamudio, A. Moscardó, C. Monteagudo, and V. Naranjo, "A self-training weakly-supervised framework for pathologist-like histopathological image analysis," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2022, pp. 3401–3405.

[33] A. E. Minarno, K. M. Ghufron, T. S. Sabrila, L. Husniah, and F. D. S. Sumadi, "CNN based autoencoder application in breast cancer image retrieval," in *Proc. Int. Seminar Intell. Technol. Appl.*, Sep. 2021, pp. 29–34.

[34] M. Kumar, R. Singh, and P. Mukherjee, "VTHSC-MIR: Vision transformer hashing with supervised contrastive learning based medical image retrieval," *Pattern Recognit. Lett.*, vol. 184, pp. 28–36, Aug. 2024.

[35] X. Liu, X. Kang, X. Nie, J. Guo, S. Wang, and Y. Yin, "Learning binary semantic embedding for large-scale breast histology image analysis," *IEEE J. Biomed. Health Informat.*, vol. 26, no. 7, pp. 3240–3250, Jul. 2022.

[36] A. Qayyum, S. M. Anwar, M. Awais, and M. Majid, "Medical image retrieval using deep convolutional neural network," *Neurocomputing*, vol. 266, pp. 8–20, Nov. 2017.

[37] M. Jiang, S. Zhang, J. Huang, L. Yang, and D. N. Metaxas, "Scalable histopathological image analysis via supervised hashing with multiple features," *Med. Image Anal.*, vol. 34, pp. 3–12, Dec. 2016.

[38] A. Mosquera-Zamudio, L. Launet, Z. Tabatabaei, R. Parra-Medina, A. Colomer, J. Oliver Moll, C. Monteagudo, E. Janssen, and V. Naranjo, "Deep learning for skin melanocytic tumors in whole-slide images: A systematic review," *Cancers*, vol. 15, no. 1, p. 42, Dec. 2022.

[39] M. Liu, Y. He, M. Wu, and C. Zeng, "Breast histopathological image classification method based on autoencoder and Siamese framework," *Information*, vol. 13, no. 3, p. 107, Feb. 2022.

[40] F. A. Spanhol, L. S. Oliveira, C. Petitjean, and L. Heutte, "A dataset for breast cancer histopathological image classification," *IEEE Trans. Biomed. Eng.*, vol. 63, no. 7, pp. 1455–1462, Jul. 2016.

[41] X. Li, X. Shen, Y. Zhou, X. Wang, and T.-Q. Li, "Classification of breast cancer histopathological images using interleaved DenseNet with SENet (IDSNet)," *PLoS ONE*, vol. 15, no. 5, May 2020, Art. no. e0232127.

[42] A. Kumar, S. K. Singh, S. Saxena, K. Lakshmanan, A. K. Sangaiah, H. Chauhan, S. Shrivastava, and R. K. Singh, "Deep feature learning for histopathological image classification of canine mammary tumors and human breast cancer," *Inf. Sci.*, vol. 508, pp. 405–421, Jan. 2020.

[43] Ü. Budak, Z. Cömert, Z. N. Rashid, A. Şengür, and M. Çıbuk, "Computer-aided diagnosis system combining FCN and bi-LSTM model for efficient breast cancer detection from histopathological images," *Appl. Soft Comput.*, vol. 85, Dec. 2019, Art. no. 105765.



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